

End Semester Examination
School of Computer and Systems Sciences
Jawaharlal Nehru University
MCA III Semester
Cloud Computing and IoT (CS -426)

December 09, 2025
MM: 50
Time: 3 hrs.

Note: Q 11 is Compulsory. Attempt any Eight (08) questions from Q1-Q10.

- Q1.** Let the availability of a compute cluster be 99.99% with a planned downtime of 52 hours. Calculate the effective availability of the system. (5)
- Q2.** Discuss the Static and Dynamic resource provisioning models for cloud computing considering the advantages and disadvantages of both. (5)
- Q3.** Discuss the CAP theorem. What is the use of Graph Database considering its applicability in Cloud Computing. (5)
- Q4.** Differentiate between the Forward Recovery and Backward Recovery schemes using check pointing. (5)
- Q5.** What is a Critical Section in program code? Discuss the methods for handling Critical Instructions for creation of an efficient VMM. (1+4=5)
- Q6.** Internet of Things (IoT) comprises of a large number of heterogeneous things. How this large number of devices can be addressed? (5)
- Q7.** Discuss the Entity based IoT Reference Model and explain its various components. (5)
- Q8.** Discuss the basic working of a Thermal Sensor and a Sound Sensor. (5)
- Q9.** Discuss the mechanism of Key –Value placement using Cassandra. (5)
- Q10.** Discuss the use of Memtable and SSTable in the context of Cassandra. What is the use of Compaction in the same context? (5)
- Q11.** Discuss the working of Scalable Weakly-consistent Infection style Membership Protocol (SWIM). Prove that it is practically almost impossible to miss a node failure using the same. (10)

***** All the Best *****